

Shubhankar Borse

Staff AI Research Scientist

Generative AI, Multimodal Models, Video Generation, and Model Efficiency.
30+ papers, 120+ patent filings and ~1,000 citations.

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Patents / Webpage
U.S. Permanent Resident (Green Card holder) · No sponsorship required

EXPERIENCE

Qualcomm AI Research, Santa Clara · *Feb 2019 – Present*

Staff AI Researcher ***Nov 2023 – Present***

Heading and Directing Research & Commercial Projects

- Headed the vision-token compression project for multimodal models, directing research and commercialization efforts across multiple teams, with demonstrations on Snapdragon phones.
- Directed a year-long, cross-functional research initiative in multi-concept personalization using Reinforcement Learning, resulting in papers at NeurIPS 2025, CVPR 2026 and a live demo at NeurIPS 2025.
- Mentored research teams working on physically grounded video generation, motion-aware video retrieval, and relighting with video diffusion.
- Lead research on parameter-efficient finetuning and drove commercial efforts for zero-shot personalization and LoRA adapters.

Sr. AI Researcher ***Nov 2021 – Nov 2023***

Cross-domain research & mobile demonstrations

- Built and demonstrated the 0.5-second Stable Diffusion running on phones.
- Led panoptic and semantic segmentation for camera, autonomous driving, and XR; built open-vocabulary panoptic segmentation showcased at CES 2024.
- Led a cross-team effort on multimodal camera–radar–LiDAR research for BEV and 3D perception presented at CVPR23 and WACV23.

Deep Learning Researcher ***Feb 2019 – Nov 2021***

Individual project ownership & algorithm development

- Owned segmentation, depth estimation, and multi-task learning projects for autonomous driving and XR, presented at CVPR21, WACV21, BMVC21.
- Demonstrated an efficient real-time 4K mobile video segmentation running on a smartphone at NeurIPS 2020 and ICIP 2021.
- Designed an on-device CNN fingerprint anti-spoofing algorithm used in commercial smartphones.

KLA-Tencor, Milpitas · *May – Aug 2018*

Machine Learning Algorithms Intern

- Developed a layer-wise interpretability metric for CNNs, integrated it into a product, and used it to improve model performance.

PUBLICATIONS

Some First-Authored Papers

- Ar2Can: An Architect and an Artist Leveraging a Canvas for Multi-Human Generation **(CVPR 2026)**
- Resolving the Identity Crisis in Text-to-Image Generation **(CVPR 2026)**
- MultiHuman-Testbench: Benchmarking Image Generation for Multiple Humans **(NeurIPS Datasets & Benchmarks 2025)**
- SubZero: Composing Subject, Style, and Action via Zero-Shot Personalization **(arXiv preprint 2025)**
- FouRA: Fourier Low-Rank Adaptation **(NeurIPS 2024)**
- PosSAM: Panoptic Open-vocabulary Segment Anything **(arXiv preprint 2024; co-first author)**
- DejaVu: Conditional Regenerative Learning to Enhance Dense Prediction **(CVPR 2023)**
- X3KD: Knowledge Distillation Across Modalities, Tasks and Stages for Multi-Camera 3D Object Detection **(CVPR 2023; co-first author)**
- X-Align: Cross-Modal Cross-View Alignment for Bird's-Eye-View Segmentation **(WACV 2023)**
- Panoptic, instance and semantic relations: A relational context encoder to enhance panoptic segmentation **(CVPR 2022)**
- HS3: Learning with Proper Task Complexity in Hierarchically Supervised Semantic Segmentation **(BMVC 2021)**
- InverseForm: A Loss Function for Structured Boundary-Aware Segmentation **(CVPR (Oral) 2021)**

Some Co-Authored Papers

- HiPhy: Hierarchical Alignment for Physically-Plausible Multi-Principle Video Generation **(arXiv preprint 2026)**
- Zero-Shot Adaptation of Parameter-Efficient Fine-Tuning in Diffusion Models **(ICML 2025)**
- LoRA-X: Bridging Foundation Models with Training-Free Cross-Model Adaptation **(ICLR 2025)**

Complete publication list on Google Scholar

PATENTS

120+ filings · 13+ granted patents

Research Areas:

- **Generative AI:** Diffusion models, zero-shot personalization, content–style composition, image editing, and blended output generation.
- **Efficient Models & Adaptation:** Fourier and sparse adapters, training-free adapter transfer, diffusion distillation, hardware-aware architectures, and test-time adaptation.
- **Computer Vision:** Panoptic and semantic segmentation, BEV perception, depth completion, point-cloud processing, and boundary-aware learning.
- **Multimodal & Sensor Fusion:** Camera–radar–LiDAR integration, occlusion handling, intermodal training, and adaptive confidence mechanisms.
- **Security & Watermarking:** Video and image watermarking, and adversarial robustness.

EDUCATION

Master of Science in Electrical and Computer Engineering – ***December 2018***

Georgia Institute of Technology, Atlanta

- **GPA** 4.0/4.0
- Specialization in Computer Vision and Machine Learning

Bachelor of Technology in Electronics and Telecommunication Engineering – ***May 2017***

Veermata Jijabai Technological Institute, Mumbai

- **GPA** 9.1/10
- Specialization in Image and Signal Processing

AWARDS

Awards & Recognition

- **NeurIPS Top Reviewer** ***2026***
- **Qualcomm Distinguished Invention Award** ***2025***
  Recognized for the invention "Regional Island Attention in Multi-Human Generation."
- **CVPR Best Demo Award — Honorable Mention** ***2022***
  Real-time, accurate, and consistent video semantic segmentation on a mobile device.
- **Qualcomm CTO Award** ***2022***
- **Qualcomm Qualstar Award** ***2019***
  On-device quantizable fingerprint anti-spoofing algorithm.

SERVICE

Reviewing & Academic Service

- **~100 Papers Peer-Reviewed**
  - **NeurIPS 2022–2026:** Reviewer.
  - **CVPR 2022–2026:** Reviewer.
  - **ICLR 2025–2026:** Reviewer.
  - **AAAI 2024–2026:** Committee member.
  - **WACV 2023–2025:** Reviewer.
  - Peer review for multiple journals.
- Workshop Chair, Omnidirectional Computer Vision (OmniCV), CVPR 2022 and 2023.
- Keynote speaker, OmniCV 2023: Detection and segmentation in BEV using near-field surround cameras.